

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 311 — Past Questions & Materials
by Renance

Course: PEE 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 300

This study pack was compiled and restructured by Renance for PEE 311 students at Federal University of Petroleum Resources, Effurun. It brings together 6 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 6 — PEE 311 - Reservoir and Fluid properties

Level 300 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
FIRST SEMESTER EXAMINATION, 2024/2025
PEE 311: RESERVOIR ROCK AND FLUID PROPERTIES. TIME ALLOWED:
2HRS:30MINS UNIT: 3
ANSWER QUESTION (2) AND ANY OTHER THREE
QUESTIONS.....DATE:20/05/2025**

QUESTION 1

- Briefly define and then explain the importance of reservoir characterization.
- State four roles of a reservoir engineer.
- As a reservoir engineer how would you determine the porosity of a new field.
- With an explicit diagram using Darcy equation for an in-compressible fluid develop an expression for the average permeability for radial flow system through serial beds.
- Given the following data, evaluate the average permeability's if the reservoir is in (i) series (ii) parallel. $r_w = 5.5$ in, $r = 2600$ ft, $r_1 = 270$ ft, $r_2 = 500$ ft, $r_3 = 1100$ ft, $r_4 = 60$ ft, $k_1 = 30$ md, $k_2 = 65$ md, $k_3 = 120$ md, $k_4 = 250$ md, $h_1 = 25$ ft, $h_2 = 60$ ft, $h_3 = 105$ ft, $h_4 = 200$ ft. (16 marks)

QUESTION 2.

A producing formation at FUPRE BOS1 consists of three strata with different thickness and rock properties as shown on the table below. Given $r_w = 0.5$ ft and $r_e = 600$ ft.

Info on Strata with different thickness/rock properties

LAYER A	LAYER B	LAYER C
$K_1 = 280$ md	$K_2 = 350$ md	$K_3 = 250$ md
$h_1 = 16$ ft	$h_2 = 22$ ft.	$h_3 = 28$ ft

As a reservoir engineer in this field, you are required to evaluate the formation to aid management decision thus:

- Calculate the average permeability
- Calculate the capacity of the formation
- Calculate the average permeability after workover operation on the well which led to the reduction of strata A permeability by 40% at a radius of 15ft, strata B by 18% at a radius of 7ft and strata C by 25% at a radius of 10ft.
- Estimate the percent of the original productivity index the well will be reduced to?
- The capacity of the damaged formation. (22 marks)

QUESTION 3.

- In a simple tabular form differentiate between the following: Drainage/Imbibitions (ii) Effective/relative permeability(iii) dynamic viscosity / Kinematic viscosity(iv) capillary pressure/wettability.
- (i) State Darcy's law of fluid flow.
- A coring program was designed to obtain core samples from OML 34, in **Evwreni ML** sand whose top and bottom of sand are 4547ft and 4560ft respectively. Table below represents a result from the routine core analysis performed on the core samples obtained from intervals

Table showing reservoir rock properties

SAMPLE	EXTENT OF SAND	POROSITY %	PERMEABILITY MD	SATURATION (%)
Em. 1	4547-4549	25.5	100	40
Em. 2	4549-4551	21.8	150	43
Em. 3.	4551-4553	20.4	84	38
Em. 4	4553-4555	22.9	95	50
Em. 5	4555-4557	20.9	145	45
Em. 6	4557-4560	24.1	203	60

- i. Calculate the weight average porosity, permeability and saturation
- ii. Estimate the stock tank oil initial in place (STOIIP) in STB, given that the drainage area = 640 acres, initial oil formation volume factor = 1.245. (16 marks)

QUESTION 4.

- (a) Determine the water saturation S_w and hydrocarbon saturation S_{hc} for each reservoir sand given in table below.

Table 4 : Resistivity values of sand samples

Sand	R_{xo}	R_{mf}	R_w	R_t
PN. 1	100	18	0.25	12
PN. 2	185	26	0.84	8
PN. 3.	145	28	0.016	2
PN. 4	200	31	0.053	5

$$S_w = [(R_{xo}/R_t)/(R_{mf}/R_w)]^{5/8}$$

- (b) State the similarities between Cores and Logs.
- (c) State the differences between Cores and Logs.
- (d) Briefly state the laboratory procedure for measuring permeability.
- (e) State four factors affecting laboratory measurements of permeability and briefly explain them. (16 marks)

QUESTION 5.

Briefly define the following terms with respect to Petroleum Engineering.

- (a) (i) Formation volume factor (ii) wettability (iii) Viscosity (iv) threshold pressure (v) Refractive index. (vi) interfacial tension (IFT)
- (b) "Rise of wetting phase in a tube varies with capillary radius" briefly explain.
- (c) State and briefly explain 4 assumptions for the use of Darcys law.
- (d) A rock 10cm long and 2cm² in cross section flow 0.0080cm³/sec of 2.5cp oil under a 1.5atm pressure drop. If the oil 100% saturate the rock, what is the rock absolute permeability? (16 marks)

Paper 2 of 6 — PEE 311 RESERVOIR ROCK AND FLUID PROPERTIES 2024I2025

Level 300 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
FIRST SEMESTER EXAMINATION, 2024/2025

PEE 311: RESERVOIR ROCK AND FLUID PROPERTIES. TIME ALLOWED:
2HRS:30MINS UNIT: 3

ANSWER QUESTION (2) AND ANY OTHER THREE
QUESTIONS.....DATE:20/05/2025

QUESTION1

- (a) Briefly define and then explain the importance of reservoir characterization.
- (b) State four roles of a reservoir engineer.
- (c). As a reservoir engineer how would you determine the porosity of a new field.
- (d). With an explicit diagram using Darcy equation for an in-compressible fluid develop an expression for the average permeability for radial flow system through serial beds.
- (e.) Given the following data, evaluate the average permeability's if the reservoir is in (i) series (ii) parallel. $r_w = 5.5$ in, $r = 2600$ ft, $r_1 = 270$ ft, $r_2 = 500$ ft, $r_3 = 1100$ ft, $r_4 = 60$ ft, $k_1 = 30$ md, $k_2 = 65$ md, $k_3 = 120$ md, $k_4 = 250$ md, $h_1 = 25$ ft, $h_2 = 60$ ft², $h_3 = 105$ ft, $h_4 = 200$ ft. (16 marks)

QUESTION 2.

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$K_1 = 280$ md	$K_2 = 350$ md	$K_3 = 250$ md
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As a reservoir engineer in this field, you are required to evaluate the formation to aid management decision thus:

- (a) Calculate the average permeability
- (b) Calculate the capacity of the formation
- (c) Calculate the average permeability after workover operation on the well which led to the reduction of strata A permeability by 40% at a radius of 15ft, strata B by 18% at a radius of 7ft and strata C by 25% at a radius of 10ft.
- (d) Estimate the percent of the original productivity index the well will be reduced to?
- (e) The capacity of the damaged formation. (22 marks)

QUESTION 3.

- (a) In a simple tabular form differentiate between the following: Drainage/Imbibitions (ii) Effective/relative permeability(iii) dynamic viscosity / Kinematic viscosity(iv) capillary pressure/wettability.

- (b) (i) State Darcy's law of fluid flow.

(C) A coring program was designed to obtain core samples from OML 34, in **Evwreni ML** sand whose top and bottom of sand are 4547ft and 4560ft respectively. Table below represents a result from the routine core analysis performed on the core samples obtained from intervals

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- Calculate the weight average porosity, permeability and saturation
- Estimate the stock tank oil initial in place (STOIIP) in STB, given that the drainage area = 640 acres, initial oil formation volume factor = 1.245. (16 marks)

QUESTION 4.

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Table 4 : Resistivity values of sand samples

Sand	R_{xo}	R_{mf}	R_w	R_t
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$$S_w = [(R_{xo}/R_t)/(R_{mf}/R_w)]^{5.8}$$

- State the similarities between Cores and Logs.
- State the differences between Cores and Logs.
- Briefly state the laboratory procedure for measuring permeability.
- State four factors affecting laboratory measurements of permeability and briefly explain them. (16 marks)

QUESTION 5.

Briefly define the following terms with respect to Petroleum Engineering.

- (i) Formation volume factor (ii) wettability (iii) Viscosity (iv) threshold pressure (v) Refractive index. (vi) interfacial tension (IFT)
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Paper 3 of 6 — PEE 311 reservoir rock & fluid properties 2025

Level 300 • First Semester • PastQuestion

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DEPARTMENT OF PETROLEUM ENGINEERING

FIRST SEMESTER EXAMINATION, 2024/2025

PEE 311: RESERVOIR ROCK AND FLUID PROPERTIES. TIME

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' LAYER A LAYER B LAYER C

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(b) Calculate the Capacity of the formation —

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ock Prope

Table showing reservoir f Hii

SAMPLE EXTENT OF porosity | re MEABILITY SATURATION |

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_Em. 3. 4551-4553 20.4 gg 1 38 pines thal

LEm. 4 4553-4555 22.9 [°C |

'Em. 5 | 4555-4557 SE ae ee

TEm. 6 14557-4560 4) ae

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Paper 4 of 6 — PEE 311 Rock and Fluid Properties

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

FIRST SEMESTER EXAMINATION, 2023/2024

PEE 311; RESERVOIR ROCK AND FLUID PROPERTIES. TIME ALLOWED: 2HRS:30MINS UNIT: 3

ANSWER QUESTION (2) AND ANY OTHER THREE QUESTIONS.....+s00+eetesees

DATE:17/05/2023

QUESTION1

(a) Briefly define an oil/gas reservoir and explain the important characteristics of a reservoir rock system.

(b) Explain the roles of a reservoir engineer.

(c). As a reservoir engineer how would you determine the permeability of a new field.

(d). With an explicit diagram, develop an expression for the average permeability for radial flow system through parallel

beds.

(c.) Given the following data, evaluate the average permeability's if it is in (i) series (ii) parallel. $r_y = 5.5$ in, $r = 2600$ ft, r_n

$= 270$ ft, $r_a = 500$ ft, $r_3 = 1100$ ft, $t_a > 60$ ft, $k_a = 30$ md, $k_o = 65$ md, $k_3 = 120$ md, $k_4 = 250$ md, $h_i = 25$ ft, $h_e = 60$ ft?, $h_s =$

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$h_i = 16\text{ft}$ —

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(b) Calculate the capacity of the formation

(c) Calculate the average permeability after workover operation on the well which led to the reduction of strata A

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PN. 2 185 26 0.84 8

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[PN.4 [200 31 0.053 5

$S_w - [(R_{xo}/R_t)/(R_{mf}/R_w)]^{**}$

(b) State the similarities between Cores and Logs.

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QUESTION 5.

Define the following terms with respect to Petroleum Engineering.

(a) (i) Formation volume factor (ii) wettability (iii) Viscosity (iv) threshold pressure (v) Refractive index.

(b) In a simple tabular form differentiate between the following: Drainage/Imbibitions (ii) Effective/relative permeability (iii) dynamic viscosity / Kinematic viscosity (iv) capillary/wettability.

(c) Consider a reservoir whose resistivity log is as shown in the fig below. The reservoir water resistivity is $1.2 \Omega\text{m}$. If the

Humble equation applies to this reservoir, and the saturation exponent is 2.2, estimate the oil saturation at 4226 ft

where the porosity is 24%. (16 marks)

Figure below: Resistivity Log.

Paper 5 of 6 — PEE 311 PAST QUESTION

Level 300 • First Semester • PastQuestion

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ROLEUM RESOURCES t .

PETERS

| FEDERAL UNIVERSITY OF TECHNOLOGY

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COLLEGE OF Fertil Gas Engine

Department of Petrol

1** Semester Examination 2017 ; :

| Course Title: Reservoir \ &

jCourse Code: PNG 311 Rim

| Instruction: Answer ALL q P

(Question 1) ;

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Sample | Thickness ft Oil ! |;

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culate (4) specific gravity, (b) the FVR,. And (c) the density of the pas mixture woder inial

servi conditions (2 marks)

mestion 3 a

Make a sketches of the following hydrocarbon properties: Bo, Rs

and Bg as a function of pressure. : - :

Draw the phase diagrams of (i) High shrinkage oil (

_ natural gas (iv) Dry natural gas. Diagrams lo be well illustrat

jetween the four types of hydrocarbons ~ , >

t

State three important applications of reservoir rock properties (20 marks)

ii) low 'shrinkage oil, (iii) wet |

edasto ~ differentiate

: Calculate the capillary pressure and capillary rise in an oil water system from the following dala: .

$p = 30^\circ$ $p_w = 1.0 \text{ gm/cm}^3$ $p_o = 0.75 \text{ gm/cm}^3$

$r = 10\text{-}\% \text{cm}$, $\text{dow cm} = 25 \text{ dynes /cm}$:(10mks)

reyz us (tg 20-75 glen

: i

ib) What do you understand by the terms' . i .

{i) Retrograde condensation (ii) Drainage process ((iii)imbibitions process(Smks)

IGuide Equations: $\epsilon = 120 [A^{0.9} - A^{1.6}] + 15 [B^{0.5} - 540]$

$|A = y_{\text{HoStyCO}_2}$, $B = \text{Mol frac .H}_2\text{S}$, $T_s = 60$ $F = 520^\circ \text{R}$

$\text{lar} = 28.96$, $R = 10.73$, $T'_{pe} = T_{rc} - \epsilon$, $P'_{pe} = 268w(\text{Cosg})$

5 Lr

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Type $+B(1-B_{he} , h = 25y_w(\text{COse}) : . _ _ \backslash$

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COURSE TITLE: RESERVOIR ROCK AND FLUID PRO! PRTIES

| COURSE CODE: PNG 311 TIME: 2 HRS -

INSTRUCTIONS: Answer Section | and ény Three (3) others

° SECTION 1 :

PRUE or FALSE ov fill dotted lines

: phases within the reservoir

it is capable of holding fluid =

in the rock to that of-total

| ' Answer all questions. TR /

| | 7 L. "Rock and fluid properties control the flow of existing pha:

4] 2. The saturation of arock is a measure of the storage capacity thal

| rd 3. Effective porosity is defined as the ratio of the total pore spaces

i volume . , . . .

iy - 4, The forces retaining the water in the Oil and Gas zones are referred to as wetabbility forces.

{ 4 "5. Connate water saturation varies with permeability, lithology and height above free water

- . 6. -Interfacial tension describes forces acting in the interface between gas and liquid a

7. Capillary forces are the result of the combined effect of surface and interfacial tension, porey

and geometry, and the wetting characteristics of the system ; :

8. Permeability measures the capacity and ability of thé formation to transmit fluid

9. The perfect gas equation of state is $PV = Z,RT$

" 10. Density is defined as a volume per unit mass of the substance

A LL. The specific gravity as the ration of the gas density to that of the air

12, Sweet gas is gas that has been cleaned with some chemicals

(3. The Isothermal gas compressibility is the change in volume per unit yolume for a unit char pressure :

ae _ s *

~ 14 CUacentrations of up to 39e UI momiydrozaroo cumpouentSianatural gasses will soriouslyd affect the accuracy of compressibility factor oe |

; 15. Processes of generating capillary pressure curve by displacement of wetting phase by non

; ig : setting phase is calledimhibition .

i ' 16, Ratio of the density of off to that of water is called

17. The highest pressure at which a bubble of gas is liberated from oil is called

18. Ratio of volume of oil (+ the gas in solution) at reservoir pressure and temperature to the volume of oi! at standard conditions is called is

19. The state of ssure and Tempe!

eat which ail intensive properties af gas and liquid phi

20, Saturated ail ceservoir is when tbe ininy

Selvulr pressure is ven the

Answer any 3 questions

I. a) Calculate the Vo

and core analysis oy

Table 1.0: Area po

lumetric Weighted porosity of the

1 the discovery well'an

reservoir for which pertinent data from log

rosity Values

d from the 2-D Seismic data are given below,

layer Depth, fi Porosity (%) _ Area (acre) -

A 1000 ~ 1010 25 . 2500

B 10-1026 15 "20UU

Cc 1026~ 1047 > 19 soo :

D 1047 ~ 1055 20. °. "1000 * :

E 1055 — 1070 21 500+

t applications of reservoir rock properties

ns 'Saturated? and 'Unsaturation

~ b) State three importan : ,

. c) Explain in Clear tern ated' with respect to a black oil : . 4

2. a) Given the following data evaluate the

$T_w = 6$ inches; $T_e = 1600$ ft; $r_w = 200$

$k_r = 22$ mD; $k_{ry} = 47$ mD; $k_{ry} = 100$ m

$h_g = 97$ ft

b) State any three factors affecting the permeability of

c) i. Define the term Effective, Absolute and Relative

ii, State any significance of any in the form

Average permeabilities if the reservoir is in series | ;

ft; $r_w = 390$ ft; $r_e = 1200$ ft; $r_y = 1750$ ft;

D; $k_r = 251$ mD; $h_r = 26$ ft; $h_o = 50$ ft; $h_y = 100$ ft

a reservoir rock

permeability

ation, accumulation and distribution of

hydrocarbon . & wee

Wye 8 i Doo. _ : . nv Sy af

3. The following measurements from core samples are available: & ig

a Sample - $\sim h_i$, ft 9% | S_0 , % j

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27 "1S OP) 7] , & 5 .

3 710 UI 79 . |

4 2.0 - 13 74 : " |

5 2.1 \4 78 :

, "Tht 19 : 75

i

olve (i) Average Porosity (ii) Average O_i and Water Saturations assuming no 2s)

b) Calculate the capillary pressure and capillary rise "in an oil-water system. from the fol}

owing j

: data; a _ 4! |

$\theta = 30^\circ$, $\rho_o = 1.0$ gm/cm³, $\rho_w = 0.75$ gm/cm³, $r_w = 10$ on $S_{ow} = 25$ dynes/cm | =f

3 c) i. How would you determine the permeability of a new field as a reservoir Engineer?

ii. Define the following fluid properties

a. Bubble point pressure. d ;

b, Capillary pressure and fluid saturation |

c, Wettability and Formation Volume Factor cb 7 & 7

Note: Guide Equations : we . r.

Note y | a 7 !

$28 \rho_o (\cos \theta) \ln \frac{r_e}{r_w} - 2 \rho_w (\cos \theta) K_e i$

$P = r \sqrt{\gamma} (P_w - P_o) / \gamma_{oo}$

_ a ae' |

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SECTION 4

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y| fs Effective Posty Du, = fore volume

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at which the first bubble % gas PHL ts Coming Out

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©) Capillary pressure : This is the difference
in Pressure existing across the . Interface Separating
two fluid Phases at the point of Contact -

| Fluid Saturation ~ This is The fraction of
Pore Space occupied by hydrocarbons.

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FEDERAL UNIVERSITY OF PETROLEUM
RESOURCES, EFFURUN

College of Technology Department of Petroleum Engineering.

Course Title: Rock and Fluid Properties

Course Code: PNG 311

\{, Instruction: Attempt any 4 questions only

jf, Time: 2. V2 hours

‘. Question 1

Ms

a. What is an oil reservoir and explain the important characteristics of a |
7 |. reservoir rock system.

P (

b. How would you determine the permeability of a new field as a reservoir
e engineer? ‘

1G. c. With clear diagrams, develop an expression for the average permeability for
linear flow through parallel beds.

f gq Give the following data, evaluate the average permeability's if it is in (i)
Series (ii) Parallel ' |

= $r_y = 5.5$ inches, $r_- = 2600$ ft, $r_i = 270$ ft, $r_2 = 500$ ft, $r_e = 1100$ ft, $r_a = |$
 2200 ft, $k_i = 30$ md, $k_2 = 65$ md, $k_3 = 120$ md, $k = 250$ md, $h_i = 25$ ft, $h_o = ,$
 $a = 60$ ft, $h_s = 105$ ft, $h_s = 200$ ft

cs Question 2

A coring program was designed to obtain core samples from Emval-ML sand
whose top and bottom of sand are 4547ft and 4560ft respectively. Table 2.0
represents a result from the routine core analysis performed on the core samples
obtained from intervals.

Table 2.0 reservoir rock properties.

[SAMPLE | EXTENT OF SAND | POROSITY | PERMEABILITY [Rano aRTON
~e. % MD (%)

| Em.1 | 4547-4549 25.5 100 | 40 |
| Em.2 " [4549-4551 . 21.8 150 [43
| Em.3 4551 - 4553 20.4 84 | 38 \
| Em.4 4553 -4555 22.7 95 | 50 \
| Em.5 4555- 4557 20.9 145 \45 \
| Em.6 4557 — 4560 24.1 203.—=C- |60 \
|

a. Calculate the weighted average porosity, permeability and saturation.

b. Estimate the stock tank oil initially in place (STOIIP) in STB, given that the
drainage area = 640 acres, initial oil formation volume factor = 1.245

bb1/STB.

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well > of 2010 psia and 75or FEBPECUvely, pry rene
rature - 0.7, H 7 Pah re er
pe rties: SPp-& " one 28-°7% and CoO,- 10%, e Cr a
; ure a wing prope d density of the gas mixture "Um, |b
pe Oe actor anc t
r ey volu
formation

diagrams of (i) High Shrinkage oit; (it) Low Shrink
pe phase a fiv) Dry Natural gas. a
he Po asa

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6 (i) leempe Porosity (ii) Average oj) ang 7m =

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33,4 UNIVERSITY OF PETROLEUM 1 GINEERING

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B.Eng DEGREE EXAMITTON DATE: 25TH APRIL, 2019

FIRST iS

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MESTER EXAMINATION, 2018/2019 \$

TIME ALLOWED: 3HRS |

PN . ES

G 311: RESERVOIR ROCK AND FLUID PROPERTY: QUESTIONS

INSTRUCTIONS: ANSWER ANY

ant characteristics of a reservoir rock system.

a new field as a Reservoir Engineer?

r the average permeability for linear flow

port

= What is an oil reservoir and explain the im

- How would you determine the permeability of

c. With clear diagrams, develop an expression f0

through parallel beds is

; s report,

d. Given the following permeability data from 4 core analyses, calculate the average Wats Fk

permeability of the reservoir.

* 7 _ _ _

Depth (ft) 4 | 6 hp

200 | 130 | 170 Piso [140 | .

Permeability (md)

222: Define the following fluid properties: , teary 2 lcah ~

i. Bubble point pressure yaa PhS

7. = [kay uk — — >)

oii Capillary pressure. ; . ree terfe

iii. Fluid saturation . =

iv. + Wettability ah

Z . ak

v Formation volume factor . We rte

sion and interfacial tension

saturation data of PNG oil reservoir is shown graphical

ons and core analysis establish the WOC at 5023 ft a

b.. Distinguish between surface ten

c. The reservoir capillary pressure —

below. Geophysical log interpretati

following additional data are available: :

Oil density = 43.5 lb/ft³, Water density = 64.1 lb/ft³, Interfacial tension = 50 dyn/cm

Calculate (a) Connate water saturation, (b) Depth to FWL; ynesion

(c) Thickness of the transition zone, (d) Depth.to reach 50% water sturation

Q3a. Explain super compressibility factor .

b. List the various ways by which Z factor can be determined. os

c. Gas reservoir has the following gas composition 2 the initial *

femnperature are 3000 psia and 80 °E vey, BSS Ne Niel Veep telnipte sue id

Calculate: i. Apparent molecular weight; ii. Specific gravity; , wt

. iii. Z- factor; iv. Density us

Component Y_j $M_{Tei} \circ R =: em$

il U ct ct re

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No 0.01 28.01 227.49 493.1

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b. Diff i

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etween effective

c. What are the methng effective permeability and effective porosity

d. A core sample is ome of measurement of porosity in the [avorp water The initial',
5 : HG,

ated with an oil (ro = 35° API), gas aU pu '= lg/cm), the wee {
er apparatus, ang , Qh OF

the sample i

ple is 250g

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increased ; gas is isplaced by ylati? 342g. th eis

to 259g. The sample is the placed in a soxhlet Ore is now ositY water nl

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volume, 95cm? ed. After dryirig the core sample, Wer FING ne. TI a

: 7 cm? is measured in a mercury pycnomet . samP ty

oil saturation, gas saturati y PY the core 2 1 |

, ion and lithology § 7A Wey }

particular temperature and pressure !5 lg/c)- we

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ja}

QSa. Distinguish between the following pairs:

- i. Imbibitions and Drainage processed reser
- ii. Under saturated reservoir 2" 1 Sat essibiity fact
ample immersed in Wate,
- iii. Gas compressibility and Gas (|
- iv. Trap and Contact angle i
ting
- v. Wetting phase fluid and Non WE para
gp _ reserved (pa 9

b. Calculate the bulk volume of a-P gratfin is:

. Give the following date: on wg caensitY rpter ig: 19/0)

Weight of ory sampe coammerse ig wale® g-grain dens! ; lc

Weight of coated 52 osilY gssurmin asa - oa

Determine the rock ' wity of

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data: 0 = 30 g |

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FWL

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Paper 6 of 6 — PEE 311

Level 300 • First Semester • Handout

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nts. Basic terms associate with Rock and fluid

Def. 1) Capillary pressure

- Definition.
- field unit
- Importance to Petroleum Engineer

Capillary pressure

Can be defined as the rise or fall of fluids in a narrow (fine bore) tubes as a result of surface tension and wetting preference.

Capillary pressure occurs when 2 immiscible fluids are in contact in a porous medium, having a discontinuity in pressure across the interface separating them.

field unit is the psi

Importance in PE

1) To understand the displacement of one fluids by another in the pores of a porous medium

2) To understand fluid saturation, history (discharge and imbibition)

2) Refractive Index.

The refractive index of a fluid is the ratio of the speed of refracted light in a vacuum or reference medium to its phase speed in the fluid under examination or it is also the ratio of the sine of the angle of incidence to the sine of the angle of refraction.

$$n = \frac{\sin i}{\sin r}$$

The field unit is a percentage % or fraction

Uses to PE

1) It is used to determine the API gravity of crude oil

	Properties of Rock	Properties of fluid
3) <u>Formation Resistivity</u> It is the measure of the formation to impede the flow of electric current. True Formation Resistivity is the resistivity of the rock when saturated partially with water. Its unit is Ohms meter (Ωm) Importance to PE To know the saturation of a formation in well logging well logging - is used to measure the depth, or the gathering of various information pertaining to that well. 2) formation resistivity is the key to hydrocarbon saturation determination.	wettability saturation Surface tension porosity permeability	viscosity capillary pressure formation resist Refractive index
4) <u>Threshold pressure</u> It is the minimum differential pressure which must be applied over the wetting and non-wetting interface before any wetting phase material can be moved from the pore space. Unit is PSI Importance 1) It is useful in locating the freewater level in pores and oil water contact (OWC) on logs 2) It is a function of the pore size		

Relative Permeability

Relative permeability is the ratio of effective permeability of a phase to a base permeability or absolute (K) permeability.

Unit

No unit b/c is in fraction.

Importance in PE

It is useful in determining residual oil saturation and connate water saturation when plotted with water saturation.

6) Effective Porosity

Is the percentage of interconnected pore space with respect to the bulk volume.

Mathematically:

$$\phi = \frac{\text{Interconnected pore volume}}{\text{bulk volume}}$$

Significance in P.E

It is used in determining the original hydrocarbon displaced volume in place.

7) Viscosity

It is defined as the internal friction that exists between fluid molecules. It can also be defined as the property of a fluid that tends to prevent it from flowing, when subjected to an applied force.

Importance to P.E

It is important in determining the mobility ratio of a fluid which in turn controls the sweep efficiency.

8) Flow Potential

It describes the velocity field as the gradient of the velocity potential.

It is a flow phenomenon where the velocity is rotational. What are hydrocarbon fluid properties? The measurable / observable attributes that can be used to adequately characterise hydrocarbon fluids.

9) Porosity

It is a measure of the storage capacity of a rock. Differentiate between the following closely associated terms (properties)

Importance
Does it give room for hydrocarbon to be stored in a rock

10) Saturation

It is defined as the mobile hydrocarbon productive zone

11) Permeability (K)

It aids fluid flow or transmissivity. A rock which is permeable can produce hydrocarbon.

1) B/w Drainage and inhibition.

2) B/w Capillary and wettability.

3) Effective permeability and Relative permeabilities

4) Dynamic Viscosity and Kinematic Viscosity

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 carbon
 the /
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 quately
 Carbon
- 1) Viscosity
 - 2) Density
 - 3) Specific gravity
 - 4) Colour
 - 5) Solubility
 - 6) Compressibility
 - 7) compressibility factor
 - 8) formation Volume factor (FVF)
 - 9) API

the
 associated.

The fluid properties changes as pressure and temperature changes.

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